$$\varepsilon = -\frac{d}{dt}(N\Phi_B) = -\frac{d}{dt}(N\Phi_m \cos\omega t)$$
$$= -N\Phi_m(-\sin\omega t)\omega$$
$$= N\Phi_m \omega \sin\omega t$$

When the coil is rotated through 90° from initial position, sin ωt = 1. Then the maximum value of induced emf is

$$\varepsilon_m = N\Phi_m\omega$$
$$\varepsilon_m = NBA\omega \qquad \text{since } \Phi_m = BA$$

Therefore, the value of induced emf at that instant is then given by

$$\varepsilon = \varepsilon_m \sin\omega t \qquad (4.28)$$

It is seen that the induced emf varies as sine function of the time angle ωt. The graph between induced emf and time angle for one rotation of coil will be a sine curve (Figure 4.26) and the emf varying in this manner is called sinusoidal emf or alternating emf.

If this alternating voltage is given to a closed circuit, a sinusoidally varying current flows in it. This current is called alternating current and is given by

$$i = I_m \sin\omega t \qquad (4.29)$$

where I_m is the maximum value of induced current.

EXAMPLE 4.15

A rectangular coil of area 70 cm^2 having 600 turns rotates about an axis perpendicular to a magnetic field of 0.4 Wb m^{-2}. If the coil completes 500 revolutions in a minute, calculate the instantaneous emf when the plane of the coil is (i) perpendicular to the field (ii) parallel to the field and (iii) inclined at 60° with the field.

Solution

$A = 70 \times 10^{-4}$m^2; $N = 600$ turns

$B = 0.4$ Wbm^{-2}; $f = 500$ rpm

The instantaneous emf is

$$\varepsilon = \varepsilon_m \sin\omega t$$

since $\varepsilon_m = N\Phi_m\omega = N(BA)(2\pi f)$

$$\varepsilon = NBA \times 2\pi f \times \sin\omega t$$

(i) When $\omega t = 0^\circ$,

$$\varepsilon = \varepsilon_m \sin 0 = 0$$

(ii) When $\omega t = 90^\circ$,

$$\varepsilon = \varepsilon_m \sin 90^\circ = NBA \times 2\pi f \times 1$$
$$= 600 \times 0.4 \times 70 \times 10^{-4} \times 2 \times \frac{22}{7} \times \left(\frac{500}{60}\right)$$
$$= 88V$$

(iii) When ωt = 90° – 60° = 30°,

$$\varepsilon = \varepsilon_m \sin 30^\circ = 88 \times \frac{1}{2} = 44\ V$$

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